Fixed square decompositions away from a singular contact

The coefficients of a Bell inequality may enter a Gram matrix, while its testing functions must stay fixed. The cancellation below achieves this separation on two complementary regions of the measurement space. It combines the two state directions before completing squares, so every testing square root has a fixed argument.

Write the source as $\mathbf{c} = (d, \lambda, n, m, C, A, D, \alpha, \beta) \in \mathbb{R}^9_+$. The compact testing coordinates satisfy

$$a^2 + b^2 = c^2 + e^2 = 1, \qquad b \geq 0.$$

For finite angles they are $(a, b) = (u, 1)/\sqrt{1+u^2}$ and $(c, e) = (v, 1)/\sqrt{1+v^2}$. The circle coordinates also include the projective endpoints. All four real state components W, X, Y, Z are free. Define the state-linear expressions

$$\begin{aligned} E &= aW - bY, & K &= aX - bZ, & J &= eE + cK, & U &= eZ - cY, \\ E_\lambda &= eW + cX, & L_A &= bX + aZ, & M &= b(cW - eX) + a(cY - eZ). \end{aligned}$$

The compact canonical Bell slack is

$$P = \lambda E_\lambda^2 + \beta L_A^2 + nX^2 + mM^2 + CE^2 + AY^2 + (D + d/2)J^2 + (\alpha + d/2)U^2 + \frac{d}{2}(L_A^2 - Z^2). \tag{1}$$

Thus $P = F_h/4$, with the compact homogeneous normalization of the canonical parent. The only negative term is explicit.

1 A fixed family on an Alice arc

Fix $0 < B < 1$ and restrict the tests to $0 \leq b \leq B$. For a source scalar τ put

$$S_\tau = \begin{pmatrix} C - \tau & \tau \\ \tau & D - \tau \end{pmatrix}, \qquad \rho = \tau - d/2, \qquad h = \sqrt{\frac{2\alpha}{2\alpha + d}}, \qquad \vartheta = \frac{h}{1+h}.$$

Set $h = 0$ if $\alpha = d = 0$, and write $P_B = B - \vartheta B^2$.

Proposition 1. *If*

$$S_\tau \succeq 0, \qquad \rho \geq 0, \qquad 4A\rho \geq d^2 P_B^2, \tag{2}$$

then P has a positive semidefinite Gram representation on the following twenty fixed testing functions:

$$\begin{aligned} \mathcal{H}_B = \big(&E_\lambda, L_A, X, M, cE, eE, J, R, ab\ell, U, b\ell, \ell, \\ &b\sqrt{1-b}\,\ell,\ b(1-b)\ell,\ cX, Y_0, bV, bY_0, \sqrt{b(B-b)}V, \sqrt{b(B-b)}Y_0\big), \end{aligned} \tag{3}$$

where $R = beX + aU$, $\ell = cY$, $V = cK$, and $Y_0 = eY$. Its matrix is affine in the nine source coordinates and three free auxiliaries τ, p, q.

Proof. Since $J - eE = cK$, the exact identity

$$CE^2 + DJ^2 = C(cE)^2 + (eE, J)S_\tau(eE, J)^T + \tau(cK)^2$$

keeps the large source coefficients in one positive block. Also $L_A^2 - Z^2 = X^2 - K^2$. Removing the first four event squares in (1), $C(cE)^2$, the S_τ block and $dJ^2/2$ leaves

$$G = \frac{d}{2}R^2 + \alpha U^2 + A\ell^2 + \frac{d}{2}b^2\ell^2 + \frac{d}{2}(cX)^2 + \rho V^2 + AY_0^2 + dbVY_0. \tag{4}$$

The coupling between V and Y_0 can be redistributed because

$$bVY_0 = aR\ell - U\ell - b^2\ell^2. \tag{5}$$

Using the fraction ϑb of this identity gives

$$G = \frac{d}{2}(R+\vartheta ab\ell)^2 + \left(\sqrt{\alpha}\,U - \sqrt{\frac{d(1-h)}{2(1+h)}}\,b\ell\right)^2 + A\ell^2 + \frac{d}{2}(cX)^2$$
$$+ \frac{dh}{(1+h)^2}(b\sqrt{1-b}\,\ell)^2 + \frac{dh^2}{2(1+h)^2}(b(1-b)\ell)^2 + \rho V^2 + AY_0^2 + d(b-\vartheta b^2)VY_0. \tag{6}$$

The relation $2\alpha(1-h^2) = dh^2$ verifies this identity, including $d = 0$ and $\alpha = 0$.

It remains to complete the last two-variable form. Its matrix has the fixed interpolation

$$\begin{pmatrix} \rho & d(b-\vartheta b^2)/2 \\ d(b-\vartheta b^2)/2 & A \end{pmatrix} = \frac{b^2}{B^2}H_+ + \frac{(B-b)^2}{B^2}H_0 + b(B-b)Q_*, \tag{7}$$

where

$$H_+ = \begin{pmatrix} \rho & dP_B/2 \\ dP_B/2 & A \end{pmatrix}, \qquad H_0 = \begin{pmatrix} \rho & 0 \\ 0 & A \end{pmatrix}, \qquad Q_* = \begin{pmatrix} 2\rho/B^2 & d/(2B) \\ d/(2B) & 2A/B^2 \end{pmatrix}.$$

Condition (2) makes H_+ and H_0 positive semidefinite. Since $0 \le \vartheta \le 1/2$, $P_B \ge B/2$, and hence $\det Q_* = 4A\rho/B^4 - d^2/(4B^2) \ge 0$ as well. All factors in (6)–(7) belong to the span (3). Their only testing radicals are $\sqrt{1-b}$ and $\sqrt{b(B-b)}$.

To obtain an affine pencil, expand the displayed Gram and replace $d\vartheta$ by p and $d\vartheta^2$ by q. For example the two square blocks on $(R, ab\ell)$ and $(U, b\ell)$ are

$$\begin{pmatrix} d/2 & p/2 \\ p/2 & q/2 \end{pmatrix}, \qquad \begin{pmatrix} \alpha & -p/2 \\ -p/2 & d/2-p \end{pmatrix}.$$

The other coefficients in (6) are $p-q$ and $q/2$. In (7), the off-diagonal of H_+ becomes $(dB-pB^2)/2$. Each coefficient is now affine in $(\mathbf{c}, \tau, p, q)$, and direct substitution of the testing functions cancels all three auxiliary coefficients. Thus

$$P = \mathcal{H}_B^T Q_{20}(\mathbf{c}, \tau, p, q)\mathcal{H}_B \quad \text{for every } \tau, p, q \in \mathbb{R}.$$

The nonlinear choices $p = d\vartheta$, $q = d\vartheta^2$ establish one positive semidefinite witness; they impose no equations on the pencil. □

2 The complementary Bob region

Fix $\delta_0 > 0$ and restrict to $|c| \ge \delta_0$. Put $\tau_0 = d/(2\delta_0^2)$. The identity

$$CE^2 + DJ^2 - \frac{d}{2}K^2 = C(cE)^2 + (eE, J)S_{\tau_0}(eE, J)^T + \tau_0(c^2-\delta_0^2)K^2$$

uses a third fixed radical. Substitution in (1) yields

$$P = \lambda E_\lambda^2 + \beta L_A^2 + (n+d/2)X^2 + mM^2 + AY^2 + (\alpha+d/2)U^2 + \frac{d}{2}J^2 + C(cE)^2 + (eE, J)S_{\tau_0}(eE, J)^T + \tau_0(c^2-\delta_0^2)K^2.$$

Consequently $S_{\tau_0} \succeq 0$ gives a source-affine Gram matrix on

$$\Psi = (E_\lambda, L_A, X, M, Y, U, J, cE, eE, \sqrt{c^2-\delta_0^2}K).$$

This matrix has order ten and no auxiliaries. Both constructions retain all state signs and are valid at their testing boundaries. Adding the nine scalar source-sign blocks gives total orders 29 and 19.

3 A uniform native source domain

The admission conditions can be verified without fixing the relative rates of the source parameters. Normalize $d = 1$ and $\lambda = \beta = 0$. Let

$$0 < H < 1, \quad 0 < z, y \le \epsilon H, \quad 0 < r < 1, \quad L = yt > 0, \quad 0 < t < a_s := 1 + zr, \quad j = a_s - t.$$

Write $B_s = H + L$, $T = B_s + zr$, $U_s = B_s - zyr$, $S = a_s + L$, and

$$\begin{aligned}\Delta &= z + (1+z)B_s + (1+z)B_s^2 + 2z(1+z)B_s r + z^3 r^2 + 2z^2 r \\ &\quad + y\{(1+z)B_s + z + z^2 r(2-r)\}, \\ G_s &= B_s + z(H + r - 1), \\ M_s &= (1+z)TH + yG_s + yzj + zrL(1+y).\end{aligned}$$

The native C2 source coefficients are

$$\begin{aligned} n &= \frac{Ta_sS - r\Delta}{2r\Delta}, & m &= \frac{z^2(1+y)(1-r)a_sS}{2G_s\Delta}, & C &= \frac{a_sM_s}{2L\Delta}, \\ A &= \frac{ya_sTS}{2H\Delta}, & D &= \frac{(1+z)TU_sS}{2yj\Delta}, & \alpha &= \frac{TM_s}{2(1-H)\Delta}. \end{aligned} \tag{8}$$

Require $n \geq 0$. For $0 < \epsilon \leq 1/100$, the other displayed denominators and coefficients are positive.

Here are bounds that remain uniform as any positive opening tends to zero. Since $L < ya_s$, setting $q_\epsilon = 1 + \epsilon + \epsilon^2$ gives

$$H \leq B_s \leq q_\epsilon H, \quad 1 \leq a_s \leq 1 + \epsilon, \quad H \leq T \leq H(1+\epsilon)^2, \quad 1 \leq S \leq (1+\epsilon)^2, \quad U_s \geq H(1-\epsilon^2).$$

In the positive polynomial for Δ, substitution of these bounds gives

$$0 \leq \Delta - H(1+H) \leq \epsilon H(3 + 7H + 2H^2) \leq 6\epsilon H(1+H).$$

For M_s, use $G_s \leq H(1+\epsilon)^2$ and $j \leq 1 + \epsilon$:

$$H^2 \leq M_s \leq H^2(1 + 4\epsilon + 7\epsilon^2 + 5\epsilon^3 + \epsilon^4) \leq H^2(1 + 5\epsilon).$$

These estimates first give $A \leq \epsilon$. For $\eta = CD/(C+D)$ the reciprocal cancels the separate small openings:

$$\begin{aligned} \frac{1}{\eta} &= 2y\Delta\left\{\frac{t}{a_sM_s} + \frac{j}{(1+z)TU_sS}\right\} \leq \frac{4\epsilon(1+6\epsilon)}{1-\epsilon} \leq 5\epsilon, \\ \frac{1}{4A\eta} &= H\Delta^2\left\{\frac{t}{a_s^2TSM_s} + \frac{j}{(1+z)a_sT^2U_sS^2}\right\} \leq (1+H)^2\frac{(1+6\epsilon)^2}{1-\epsilon}. \end{aligned}$$

As $(1+6\epsilon)^2/(1-\epsilon) \leq 1 + 14\epsilon$, it follows that

$$\eta \geq \frac{1}{5\epsilon}, \qquad 4A\eta \geq (1+H)^{-2} - 14\epsilon.$$

The square root entering the Gram witness is equally well controlled. Put $\nu = TM_s/H^3$ and $\omega = \Delta/[H(1+H)]$. Then $1 \leq \nu \leq 1 + 8\epsilon$, $1 \leq \omega \leq 1 + 6\epsilon$, and

$$\frac{h^2}{H^2} = \frac{\nu}{H^2\nu + (1-H^2)\omega}.$$

Thus $H(1-3\epsilon) \leq h \leq H(1+4\epsilon)$, so $|h - H| \leq 4\epsilon$.

Now choose, once and for all,

$$\delta_0 = 2^{-26}, \qquad B = 1 - \delta_0, \qquad \epsilon = \frac{\delta_0^2}{4096} = 2^{-64}.$$

Set $\tau = \eta - 1/4$ and $\rho = \eta - 3/4$. For positive C, D, $S_\tau \succeq 0$ whenever $0 \leq \tau \leq \eta$; indeed its determinant is $CD - \tau(C+D)$. The bound on η ensures this condition, $\rho > 0$, and also $S_{\tau_0} \succeq 0$ for $\tau_0 = 1/(2\delta_0^2)$.

Finally, $h/(1+h)$ is 1-Lipschitz on $[0,1]$. Replacing h by H changes P_B^2 by at most 8ϵ. Therefore

$$\begin{aligned} 4A\rho - P_B^2 &\geq (1+H)^{-2} - \left(B - \frac{H}{1+H}B^2\right)^2 - 25\epsilon \\ &\geq \frac{\delta_0^2}{4} - 25\epsilon > \frac{\delta_0^2}{8}. \end{aligned}$$

The second inequality uses $B(1 + H\delta_0) \leq 1 - \delta_0^2$ and $(1+H)^2 \leq 4$. This proves both regional certificates on the entire displayed source envelope. Arbitrary nonnegative λ, β may then be restored as their whole event squares. Literal $n = 0$ is included. The rational parametrization (8) concerns positive openings; undefined zero denominators are not assigned limiting values.

4 Why the contact requires a different completion

At an exact contact, every square in a positive decomposition must vanish. For the native C2 contact parametrization, let $z = u_c^2 > 0$, $y = v_c^2 > 0$, $0 < H < 1$ and $L = yt > 0$. Its contact is

$$(u_c, v_c) = (\sqrt{z}, \sqrt{y}), \qquad (W_c, X_c, Y_c, Z_c) = \left(\frac{H+L}{\sqrt{zy}}, -\sqrt{z}\, r, \frac{H}{\sqrt{y}}, 1 \right).$$

The native source equations give $P = 0$ there, whereas

$$cE = \frac{L}{\sqrt{(1+z)(1+y)}} > 0.$$

For the source envelope above the contact lies in the omitted cap, since

$$b_c = (1+z)^{-1/2} \geq 1 - z/2 > B, \qquad |c_c| = \sqrt{\frac{y}{1+y}} < \frac{\delta_0}{64}.$$

For $C > 0$, both displayed decompositions retain the strictly positive term $C(cE)^2$. They therefore cannot remain positive semidefinite through that contact. A completion covering it must redistribute this term with the remaining state directions.

In general, if $P(t) = \Phi(t)^T Q \Phi(t)$ with $Q \succeq 0$ and $P(t_c) = 0$, then $Q\Phi(t_c) = 0$. This follows from $P(t_c) = \|Q^{1/2}\Phi(t_c)\|^2$ and is a necessary constraint on any replacement. It keeps the exact contact in the construction before positivity is proved.